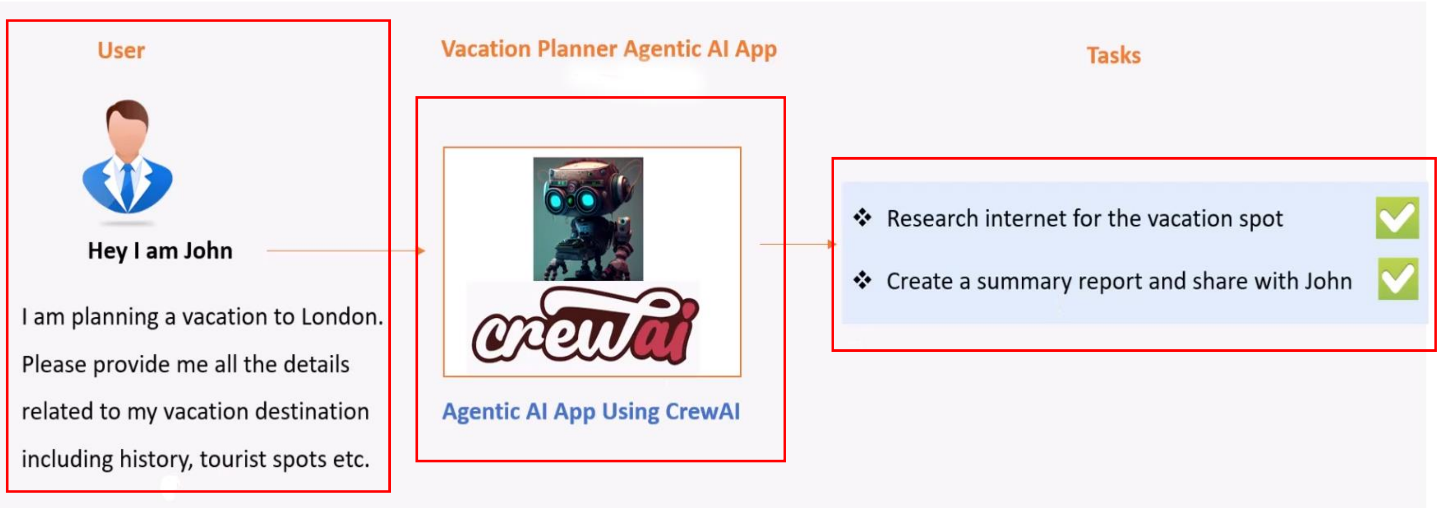
Vacation Planner Agent with crew AI

Summary: -

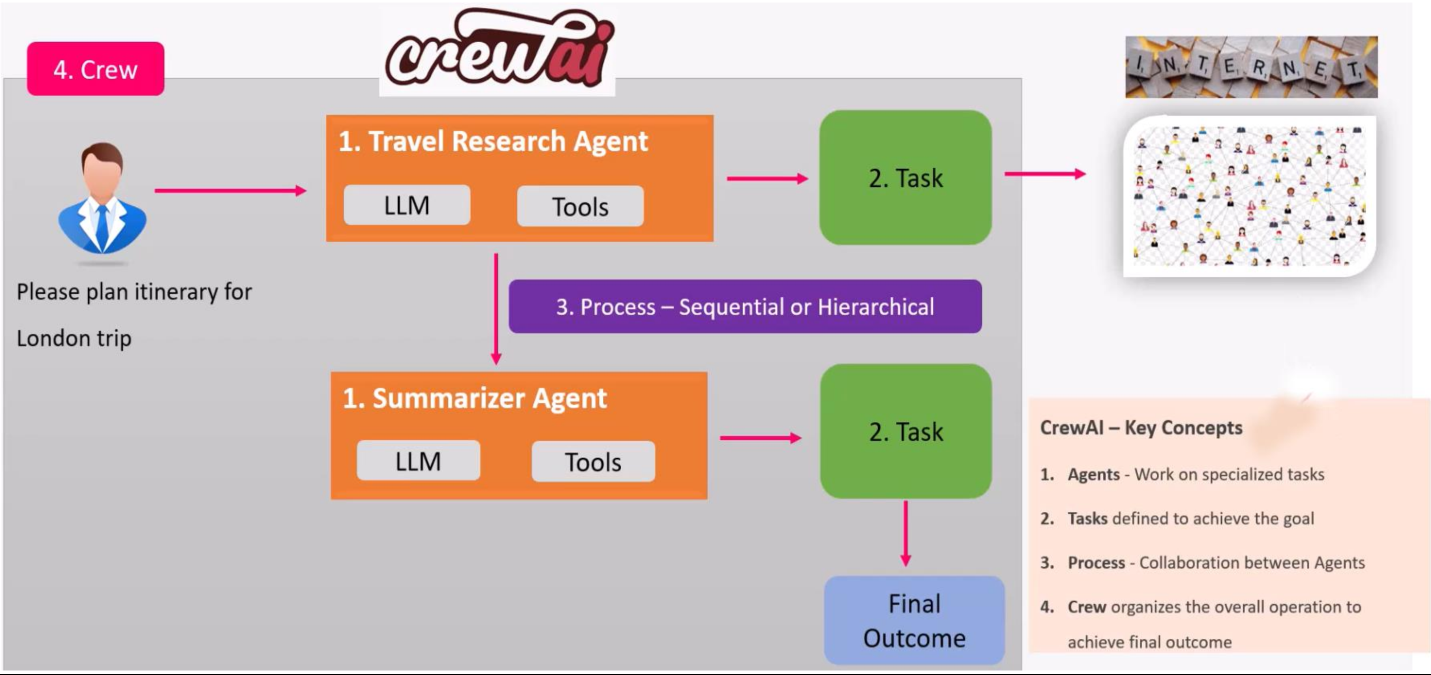
I developed an AI-powered Vacation Planner application using CrewAI, where multiple agents collaborate to generate travel plans. The system includes a research agent that gathers destination information using web tools (Serper API) and a planning agent that creates a detailed itinerary report. I configured AWS Bedrock for LLM integration, set up environment variables and dependencies, and implemented the project structure using Python and YAML configurations. Finally, I built a user-friendly interface using Streamlit to allow users to input destinations and receive complete vacation plans automatically.

Key Concept: -

- **Crew AI: -**
  CrewAI is an open-source Python framework that lets you create multiple AI agents, assign them roles, and make them work together as a "crew"
- Now let's continue with our vacation planning theme.



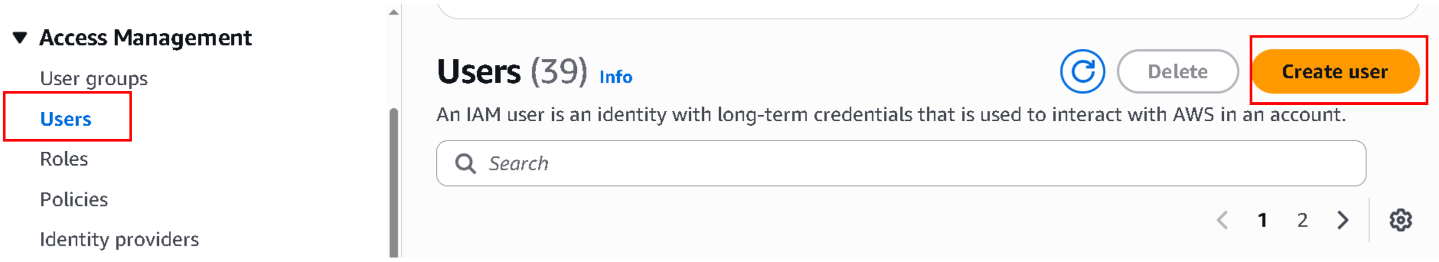
- And let's see we have a user, john.
- And john is planning a vacation to London, and he needs all the details related to the vacation destination, including history, tourist spots, and so on.
- Basically he wants a travel information for London.
- Now john can send this request to a vacation planner Agentic AI app built using Crew AI.
- Now this app will have two Agent.
- The first Agent should be able to research the internet for the vacation spot and in this case London.
- And the second Agent create a summary report and share with john.



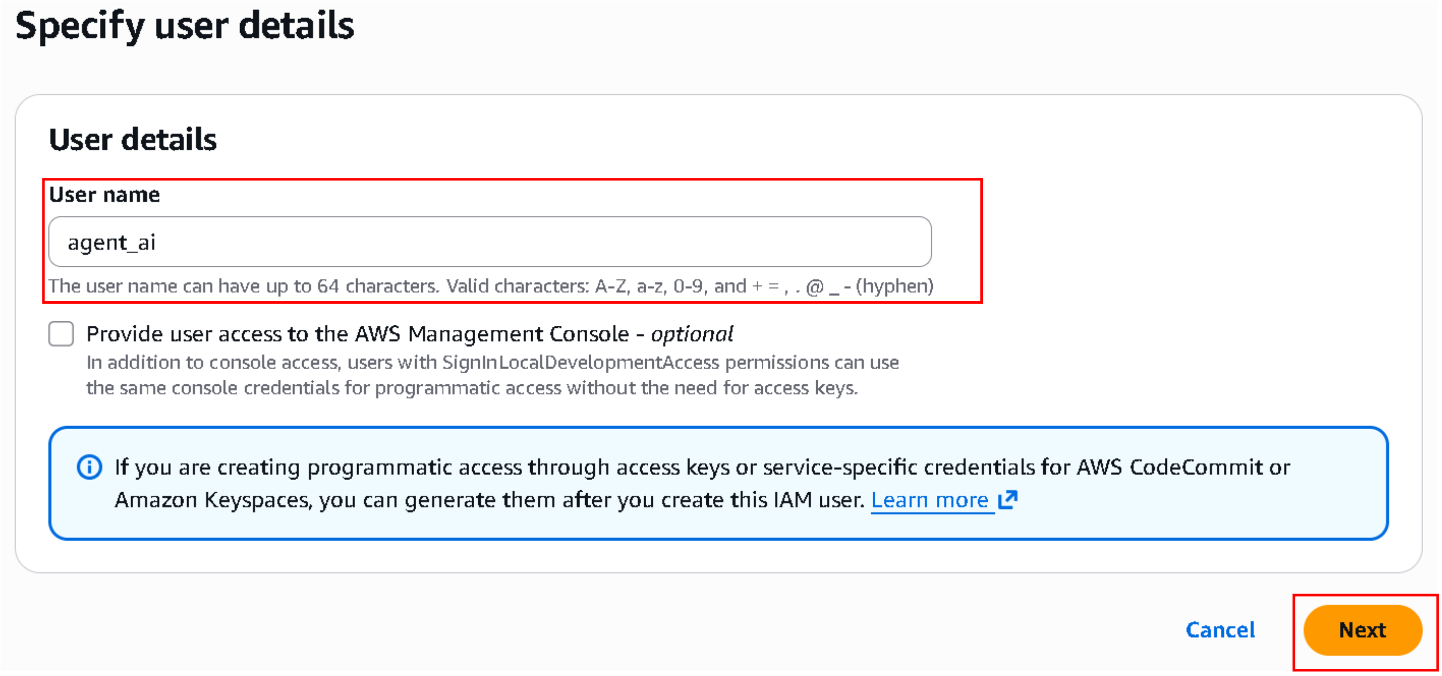
- This is the work flow of our Agents.

<u>Some PreRequisites before starting Creating App: -</u>

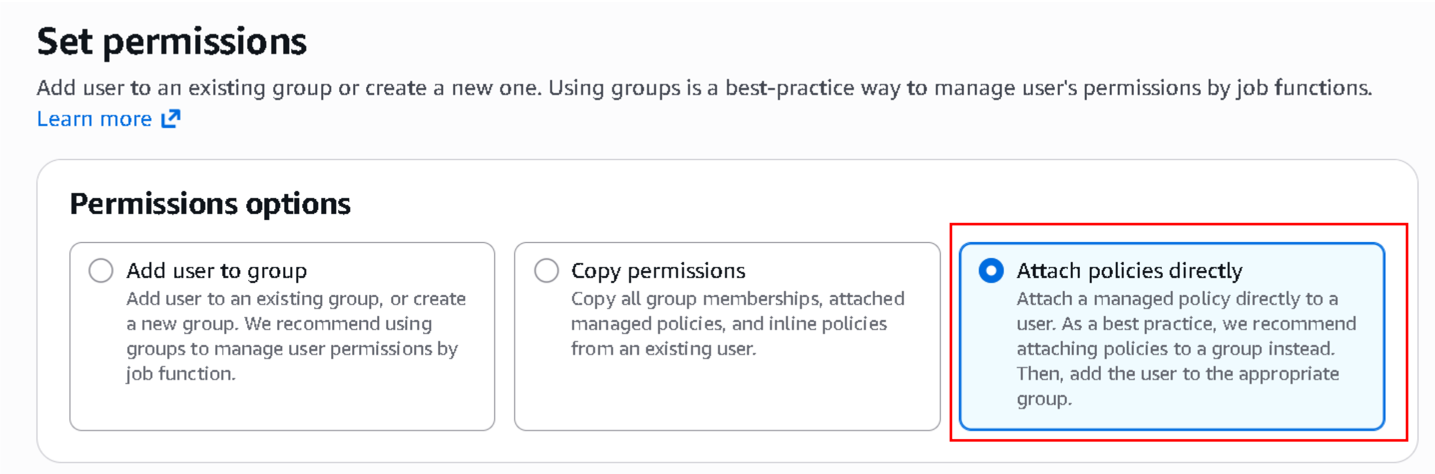
- Install Vs code: - [Link](windows)
- Install python(Recommended 3,11.9): - Link(windows)
- Install AWS CLI: - Link
- Now open Vs code and configure IAM user.
- So before you go to your VS Code open AWS console in your Browser.
- And search IAM and open it.



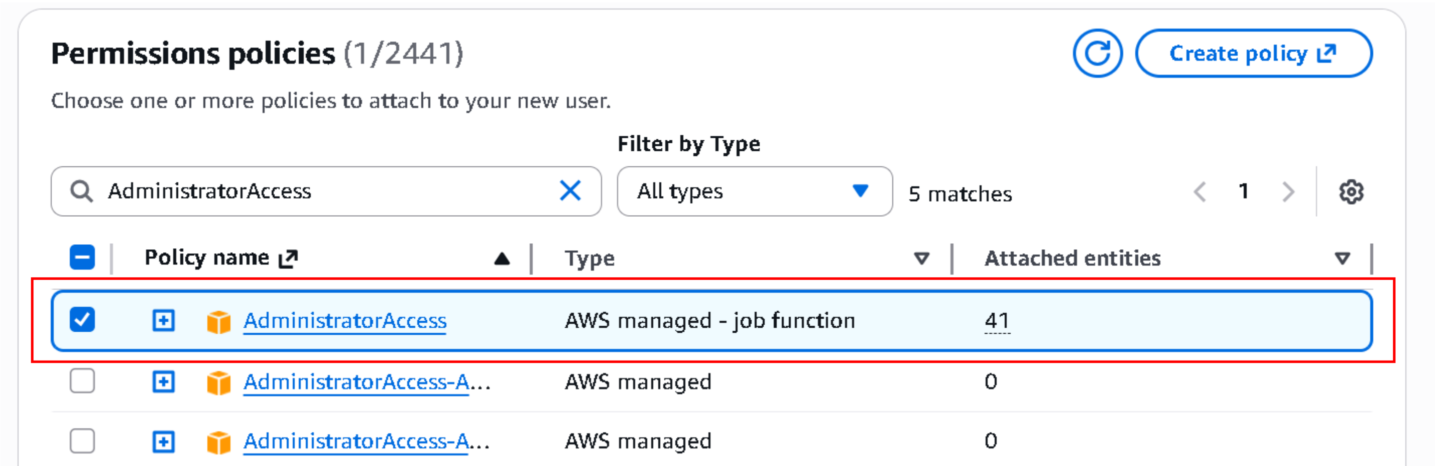
- Now in Navigation bar of IAM under Access Management click User.
- Then click Create User.



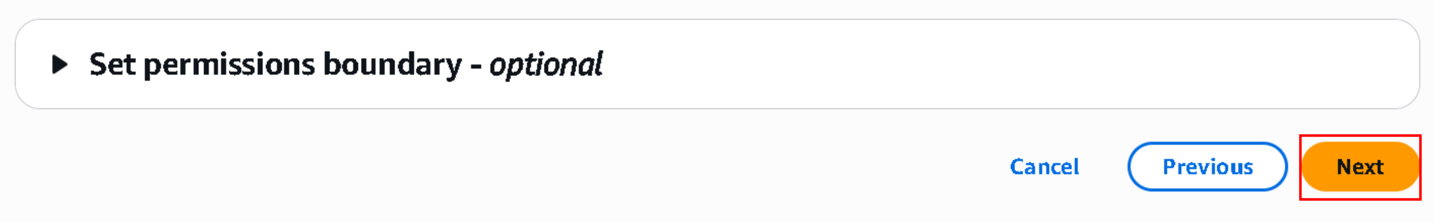
- Now Enter User name.
- And not tick the box and simply click Next.



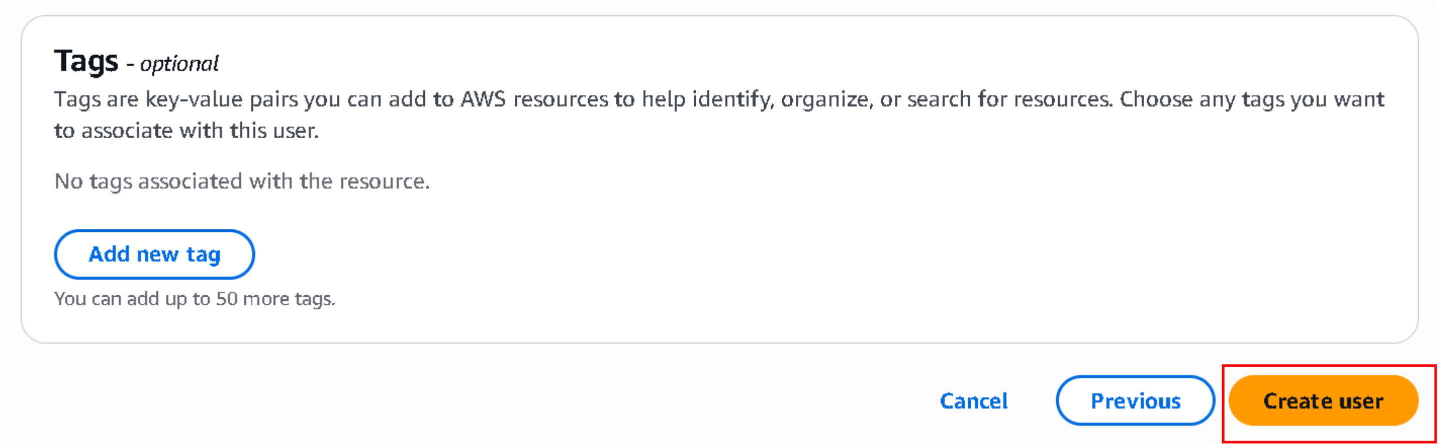
- Now in Set permissions select Attach policies directly.



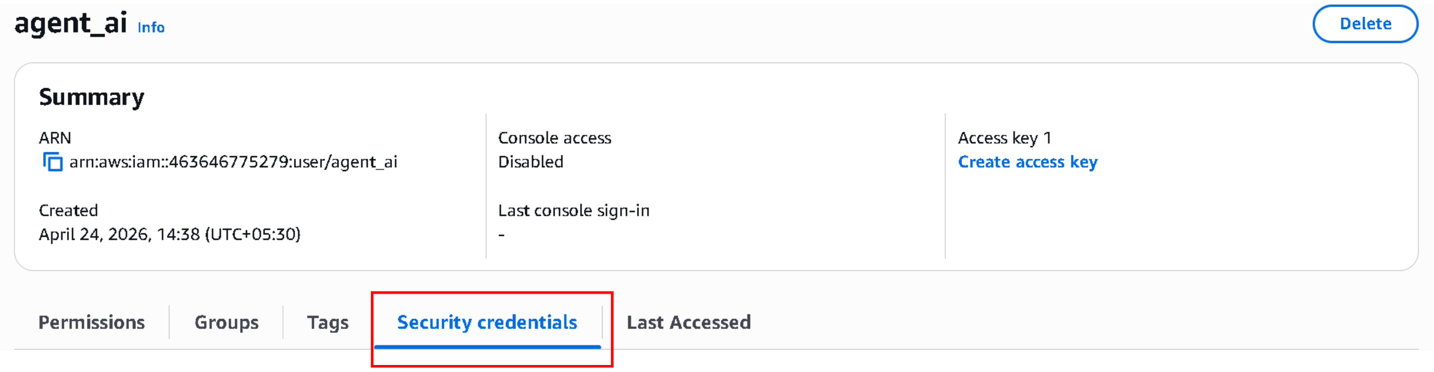
- Now in Permission Policies search and select AdministratorAccess.
- Now scroll down.



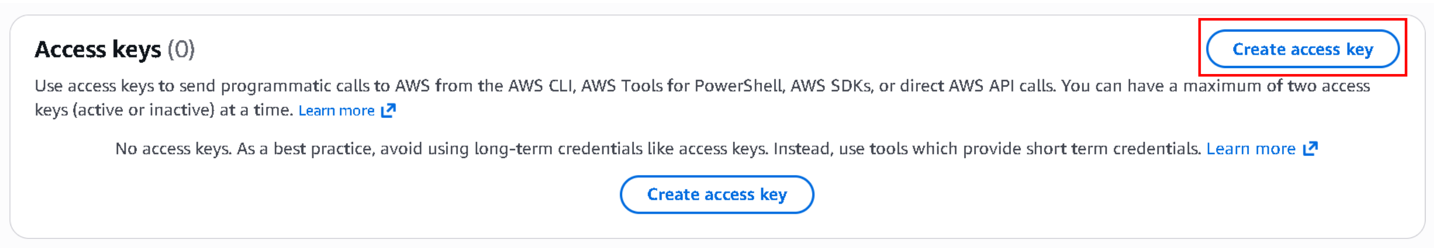
- Now simply click next.



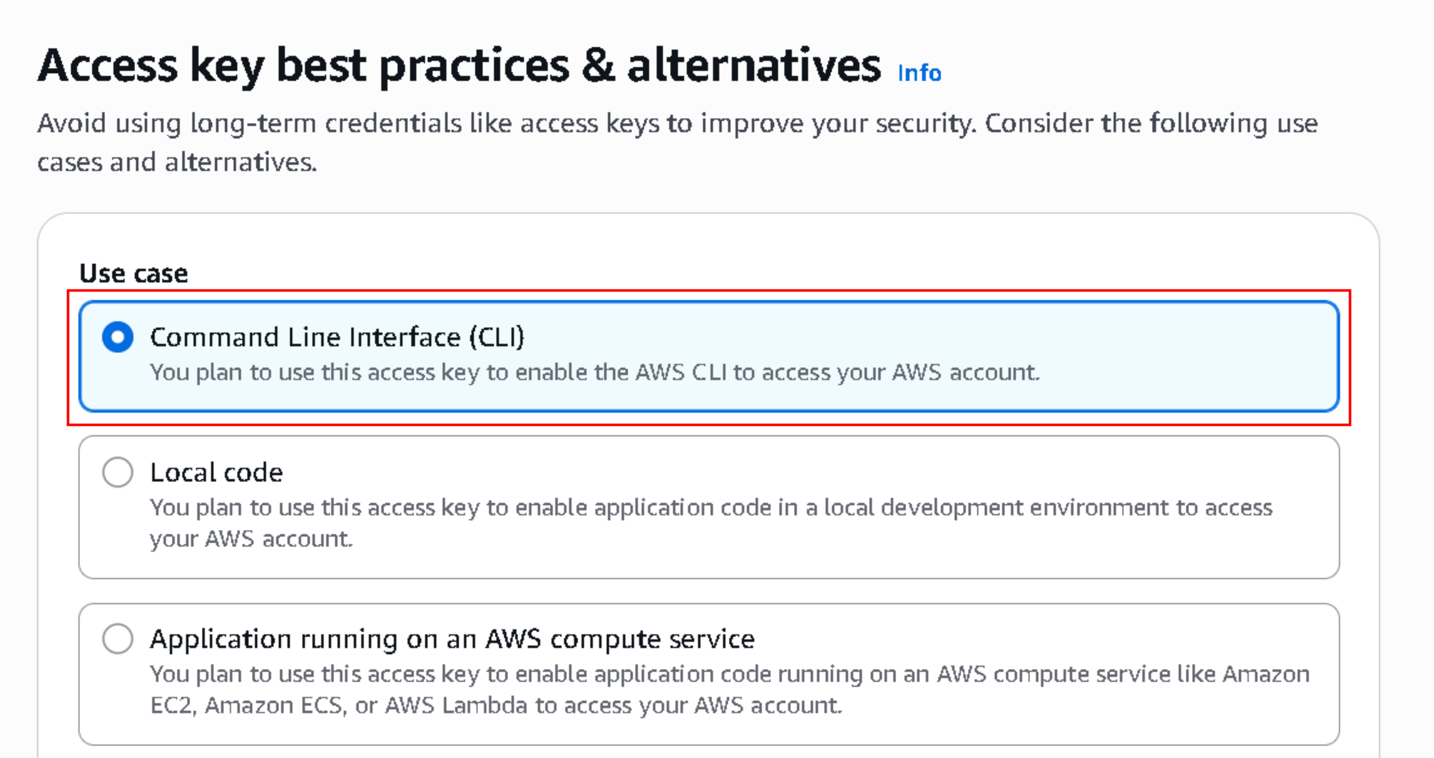
- Now review all thing and simply scroll down.
- And click Create user.



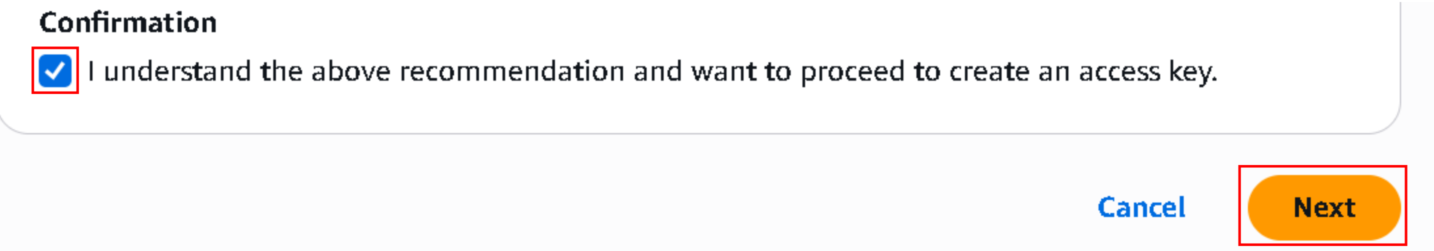
- Now our user is created.
- Now go in Security credentials.



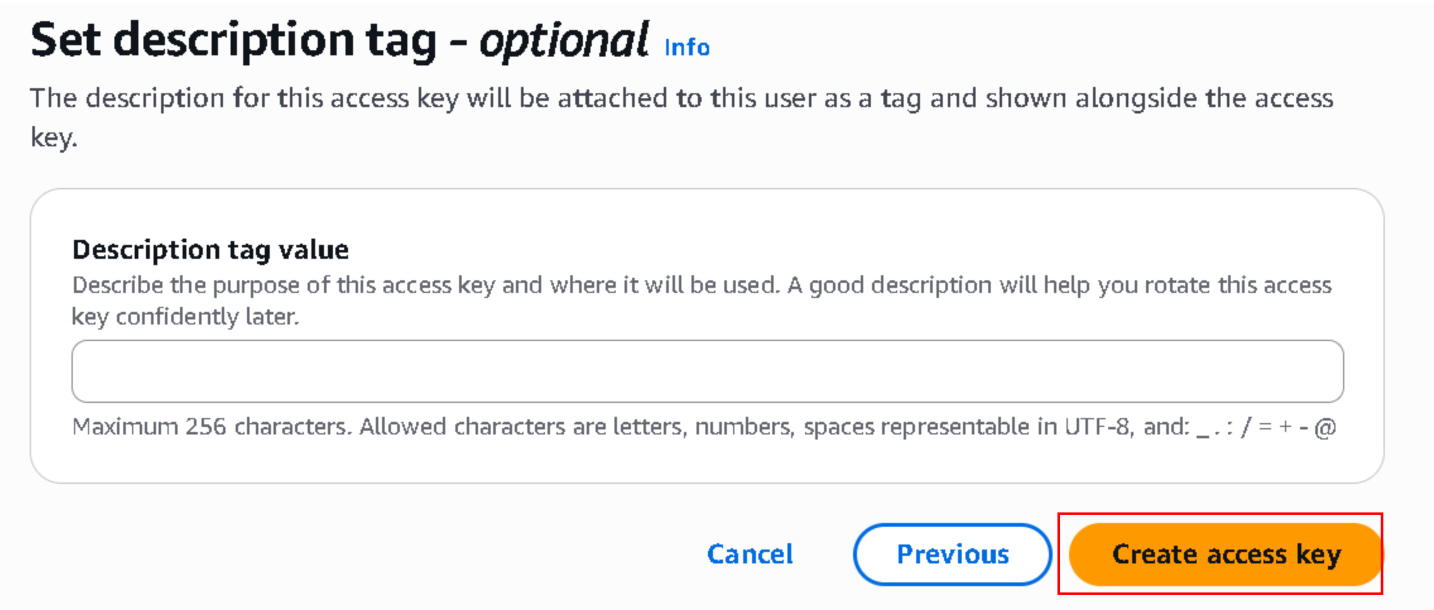
- Now simply scroll down and search Access keys.
- Now in access keys click Create access key.



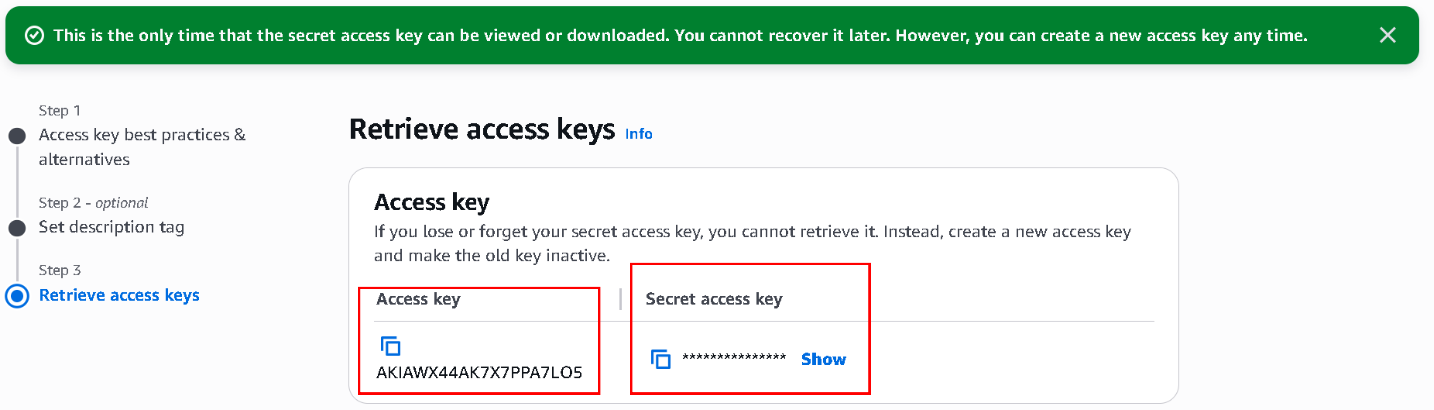
- Now simply select use case as CLI
- Then simply scroll down.



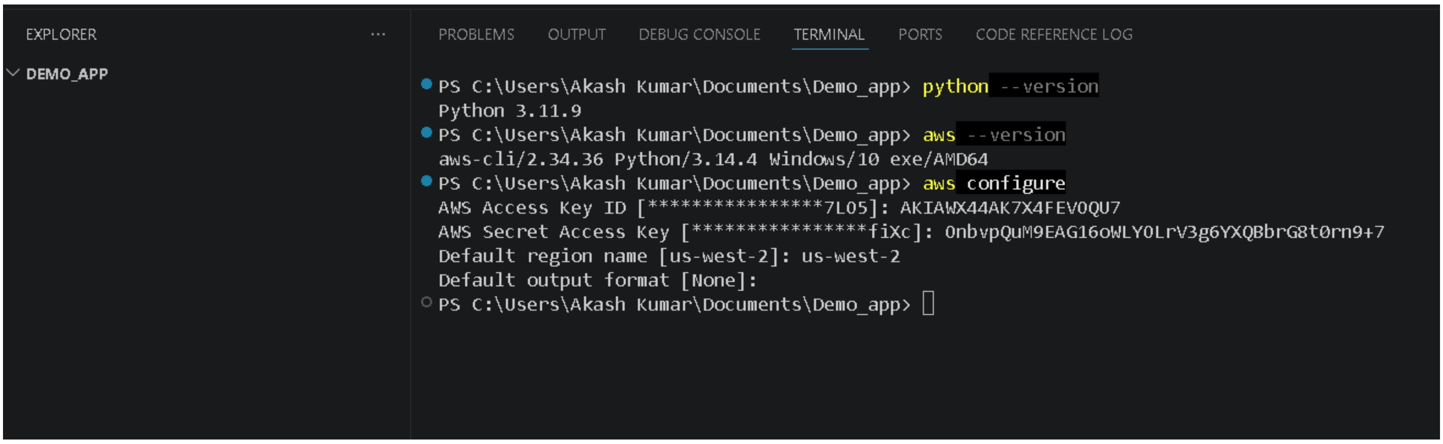
- Now tick the confirmation box and simply click Next.



- Now Simply click Create Access Key.



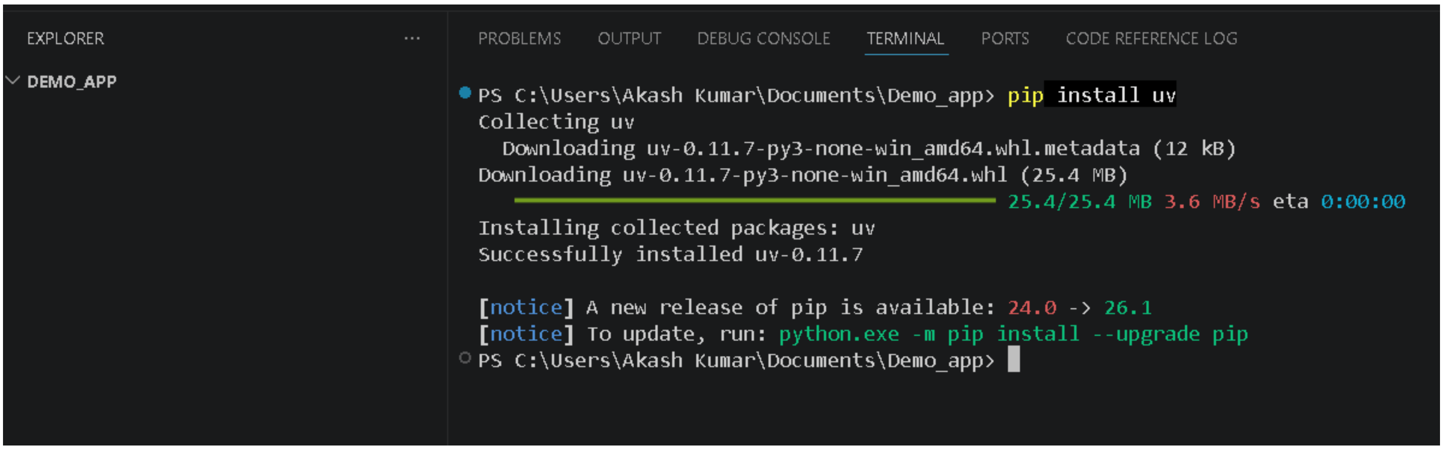
- Our Access Key is created.
- Now go to your VS Code that you open by Anaconda.



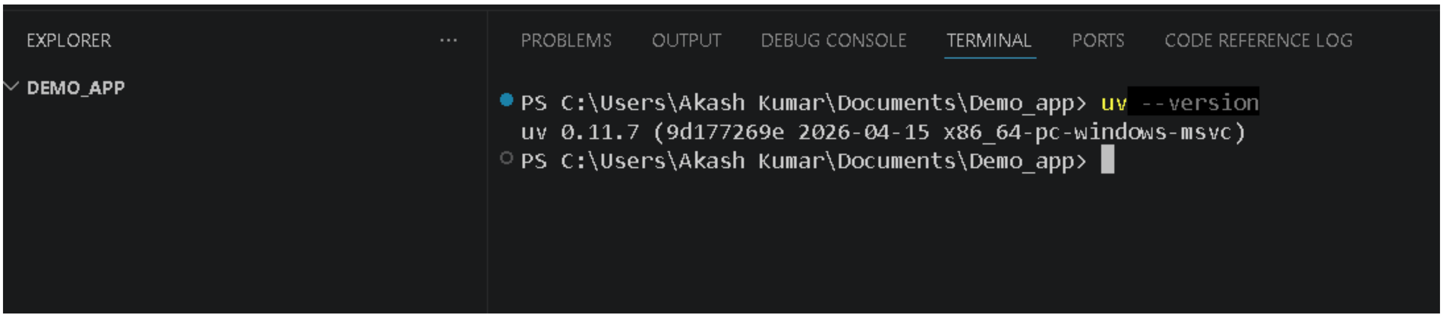
- Now firstly check python is install correctly or not by python --version
- Now check AWS cli is properly installed by aws --version
- Now all thing done then configure your Aws cli role in your VS Code
- Now write aws configure.
- Then they ask Access Key (copy access key from user that you created and paste it) then click Enter
- Now ask Secret Access Key (copy secret access key from user that you created and paste it) then click Enter.
- Then they ask region name: - give region in which you perform your all agent work
- In my case us-west-2
- Now they ask output format leave it and click Enter.
- Now configuration is done.

CrewAI Installation: -

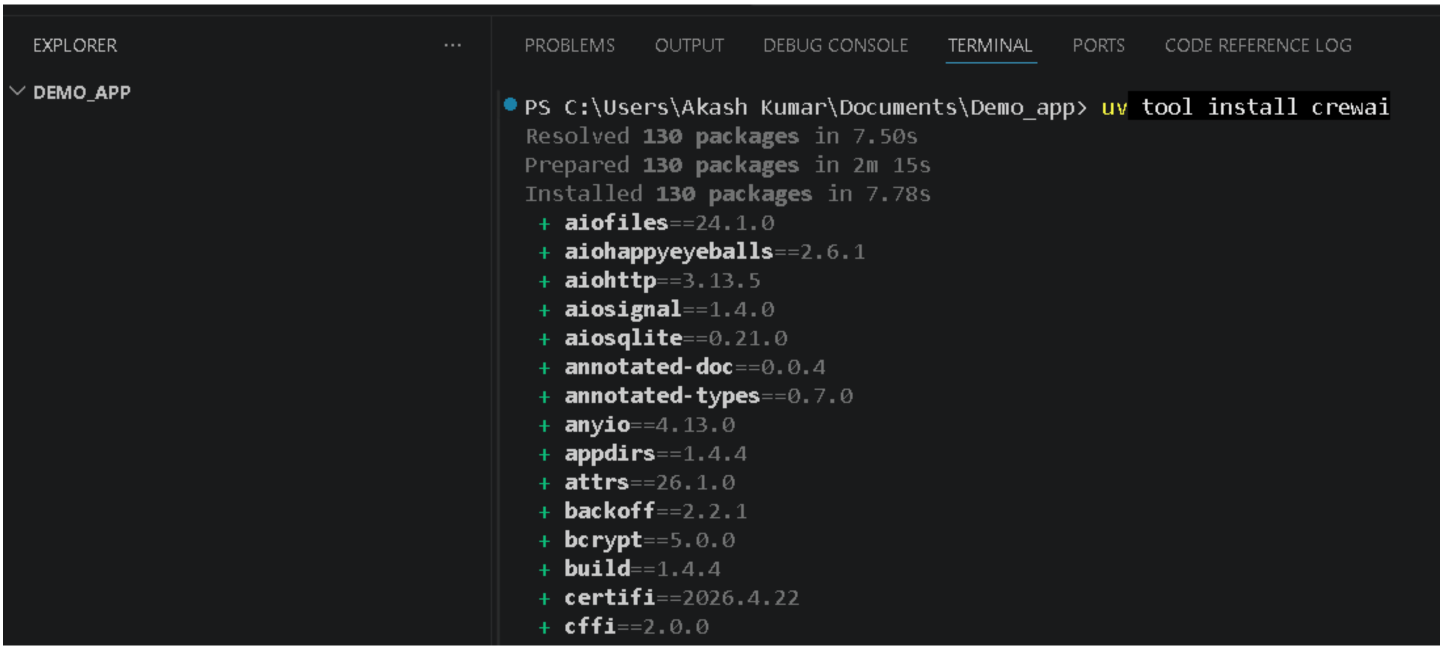
- pip install uv



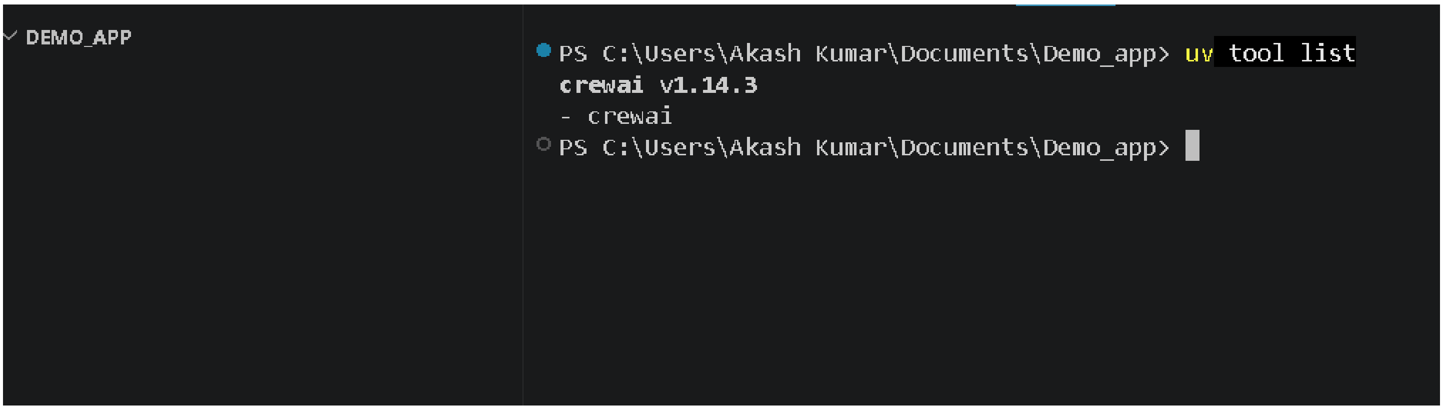
- uv –version



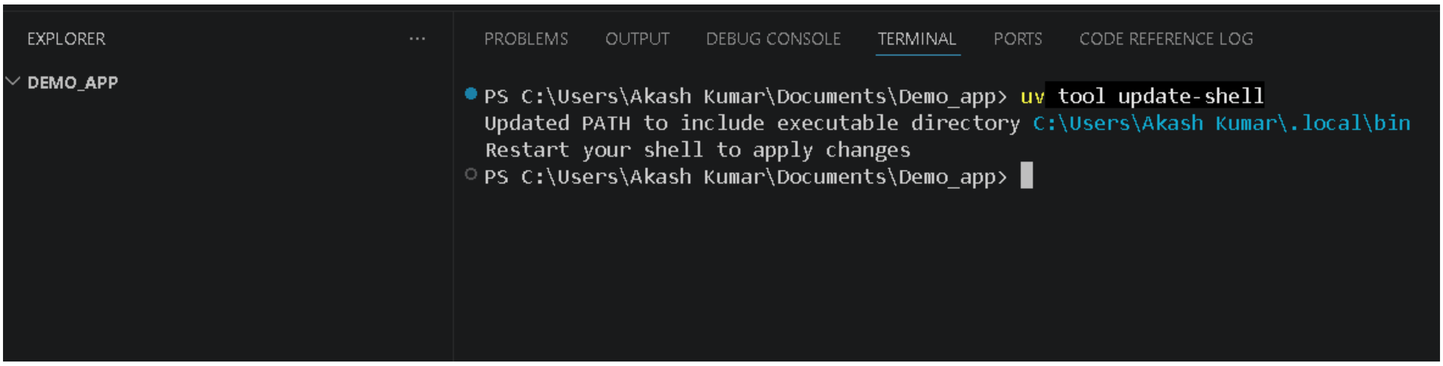
- uv tool install crewai



- uv tool list

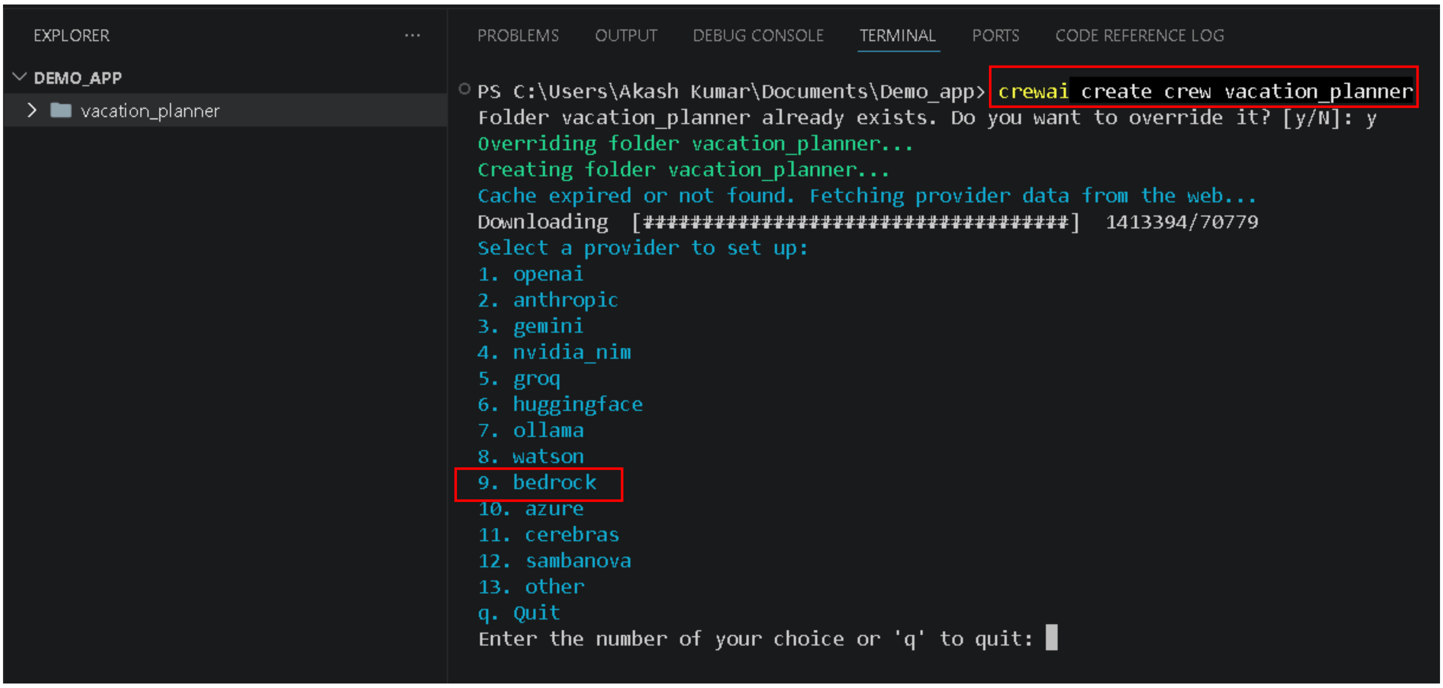


- uv tool update-shell (One time to set the path)

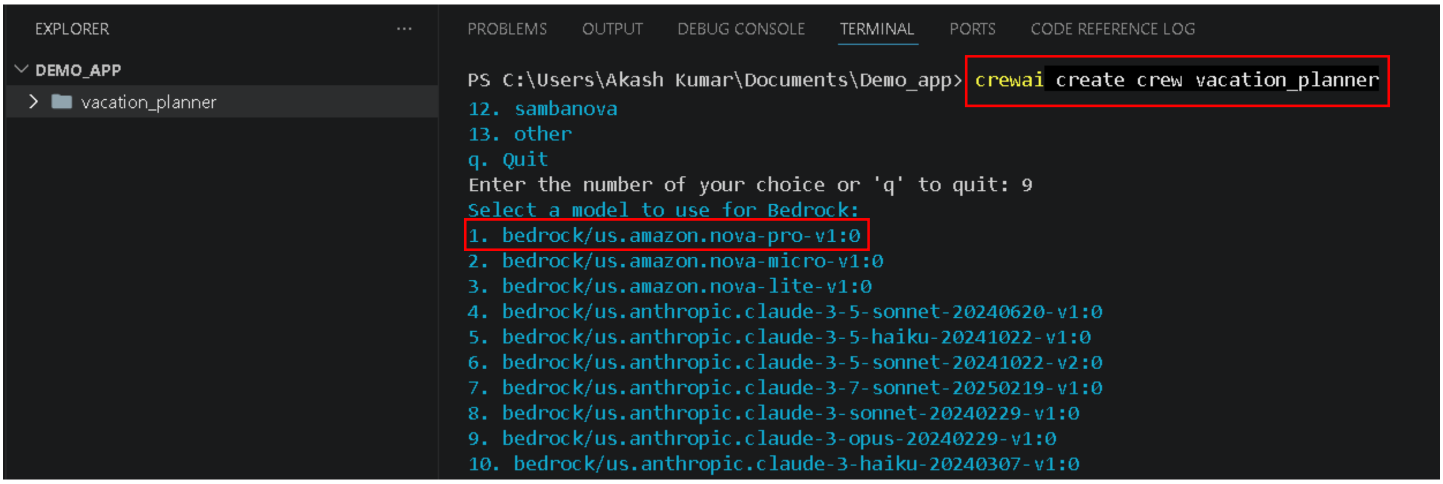


Now Create Project: -

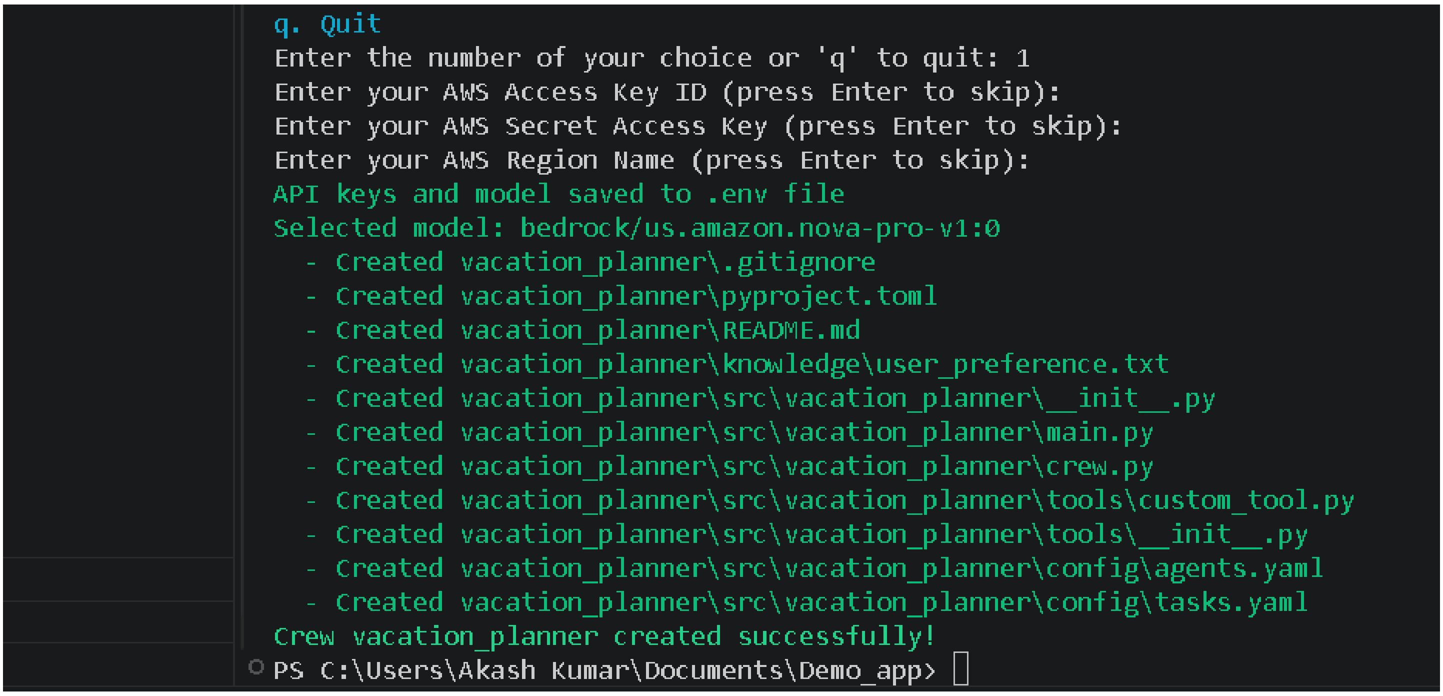
- crewai create crew vacation_planner



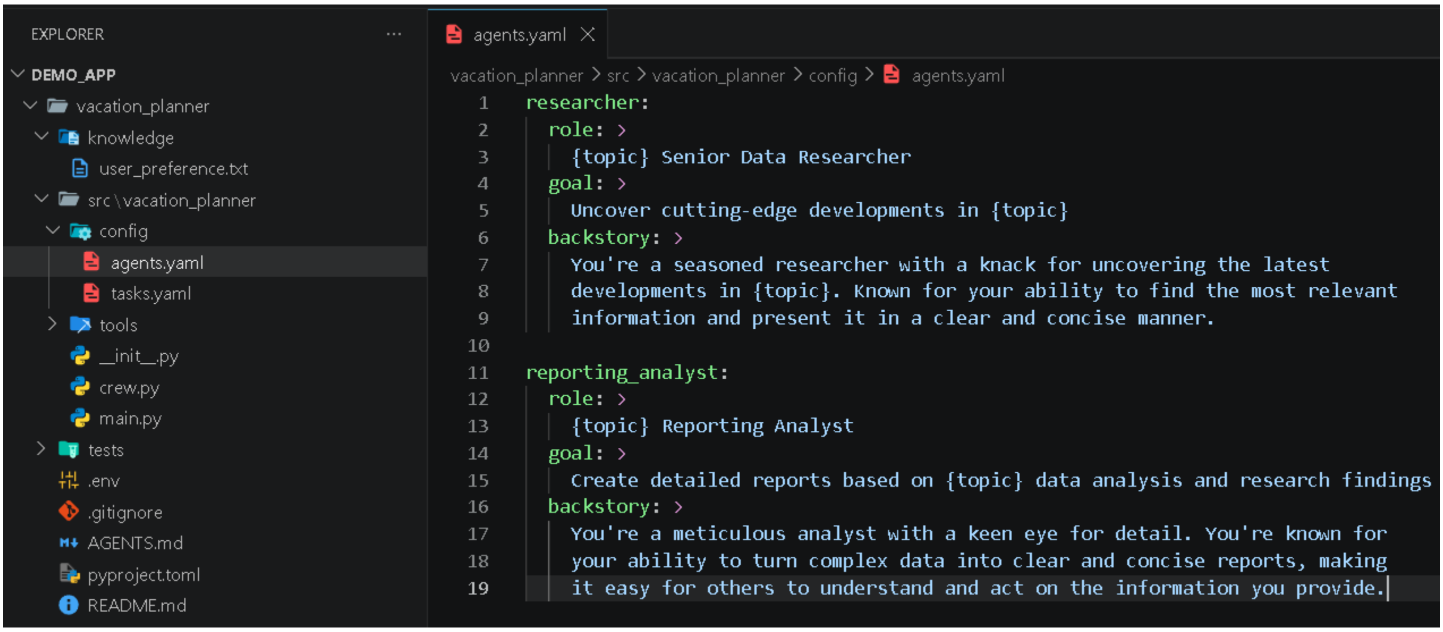
- select 9(bedrock)



- Now select amazon.nova.pro
- So scroll down and select 1.



- Now you get option to Enter Access Key, secret Access Key, Region.
- Simply Press Enter and Skip this part because we already configure our AWS cli in starting.
- Now you see in this Green it's created basically it's called project scaffolding.



- Now from Previous commands some files is created in your folder DEMO_APP like vacation_planner.
- Now go inside it and then go inside src.
- In src go inside config and open agent.yaml file .
- Now replace previous written YAML code from this code.

```
# Configuration for the agents used in the vacation planner project.

vacation_researcher:

role: >

Senior Vacation Researcher

goal: >
```

Research and find the best vacation spots related to {topic}.

backstory: >

You are a seasoned vacation researcher, known for your ability to discover hidden gems and unique experiences.

Your goal is to find unforgettable destinations tailored to a variety of travel interests.

itinerary_planner:

role: >

Itinerary Planner

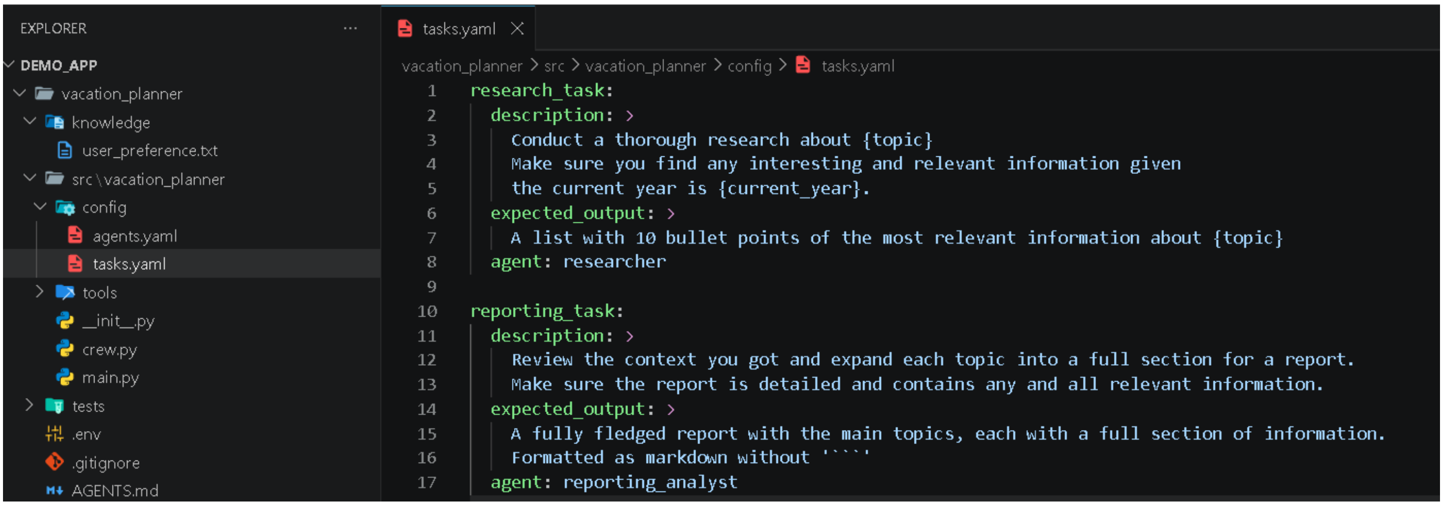
goal: >

Create a detailed itinerary highlighting the best tourist spots in {topic}, including history of the city, list of must-try local foods.

backstory: >

You are a meticulous planner with a keen eye for detail.

You specialize in crafting organized travel schedules and suggesting local culinary experiences to make vacations truly memorable.



- Now open the task.yaml present in same folder.
- And replace the previous code with this.

Configuration for tasks in the vacation planner project.

research_task:

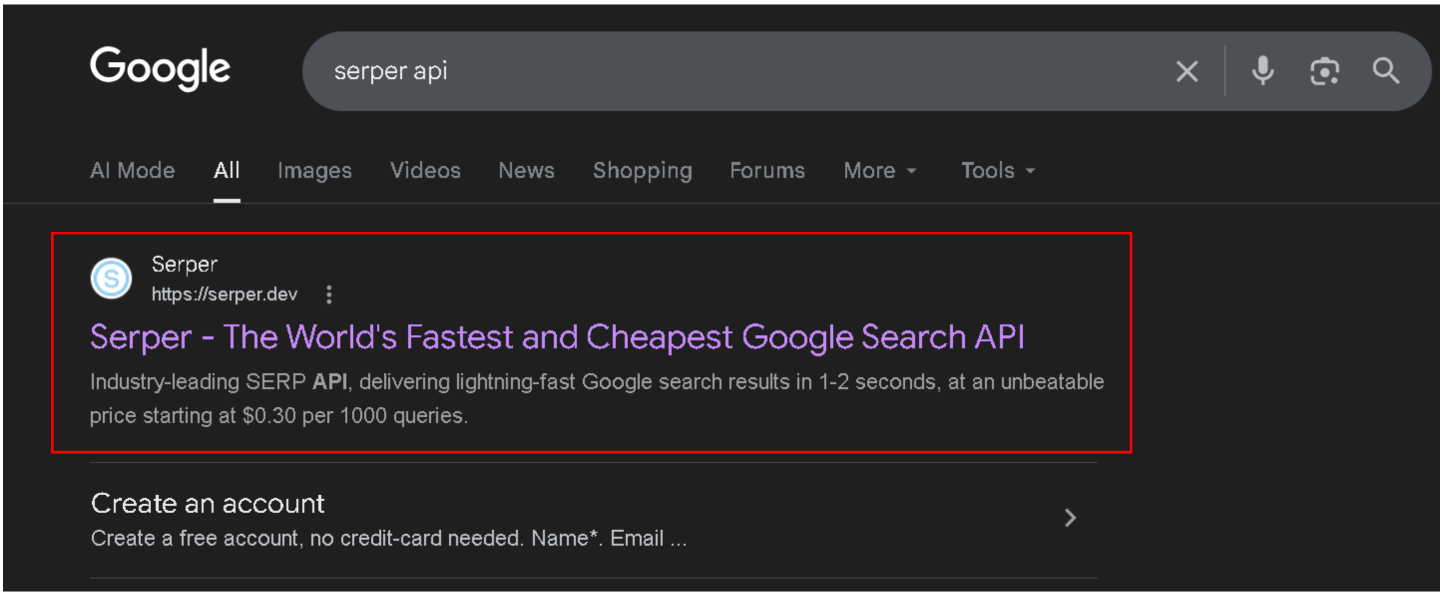
description: >

Conduct thorough research about the travel destination {topic}.

Find interesting and relevant information to enrich the travel experience.

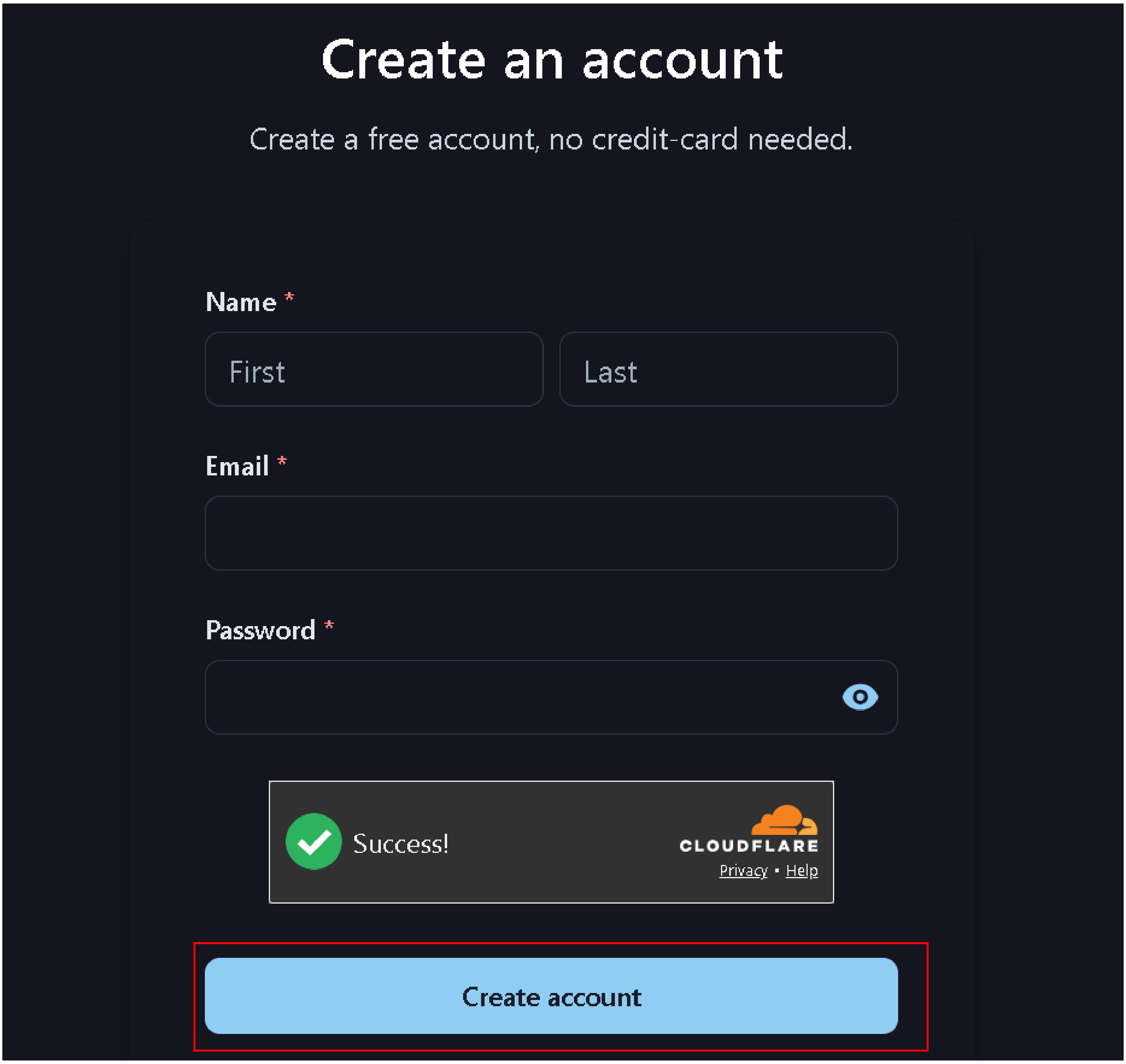
```yaml
expected_output: >

A list of 15 bullet points highlighting the most important and exciting facts about {topic}.

agent: vacation_researcher

reporting_task:

description: >

Review the research findings and create a detailed report about {topic}.

expected_output: >

A fully fledged report summarizing key information about {topic},

formatted in Markdown without using code block syntax (do not use triple backticks).

agent: itinerary_planner

output_file: report.md

output_format: markdown
```

- Add Tools to Agents in Crew.py
- Now we provide our agent, researcher agent tool which allow it to search the internet.
- So there are lot of tools but we use Google serper search.
- And for this you need a API key.
- So just we need to go on the Browser and search serper Api.

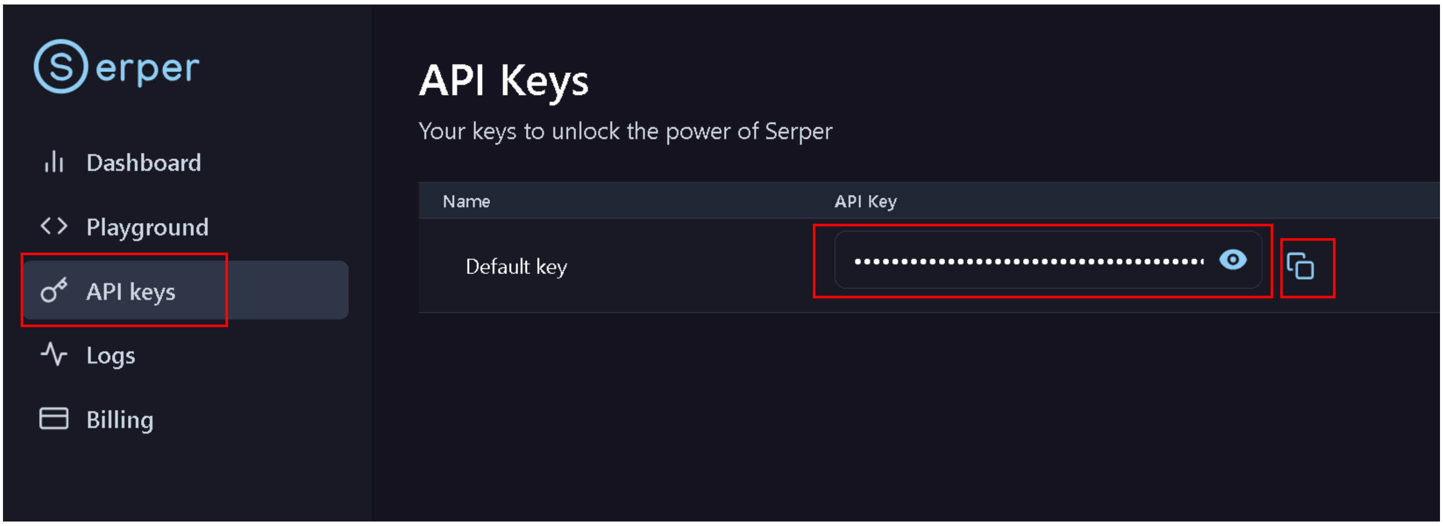


- Click on first Link.

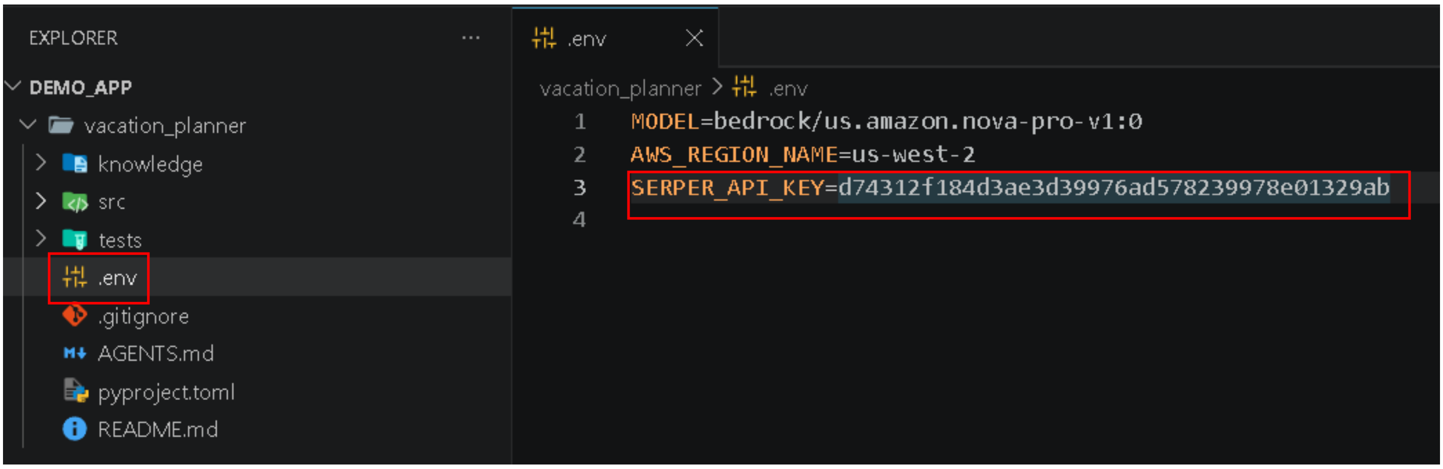
- Simply click on signup button.



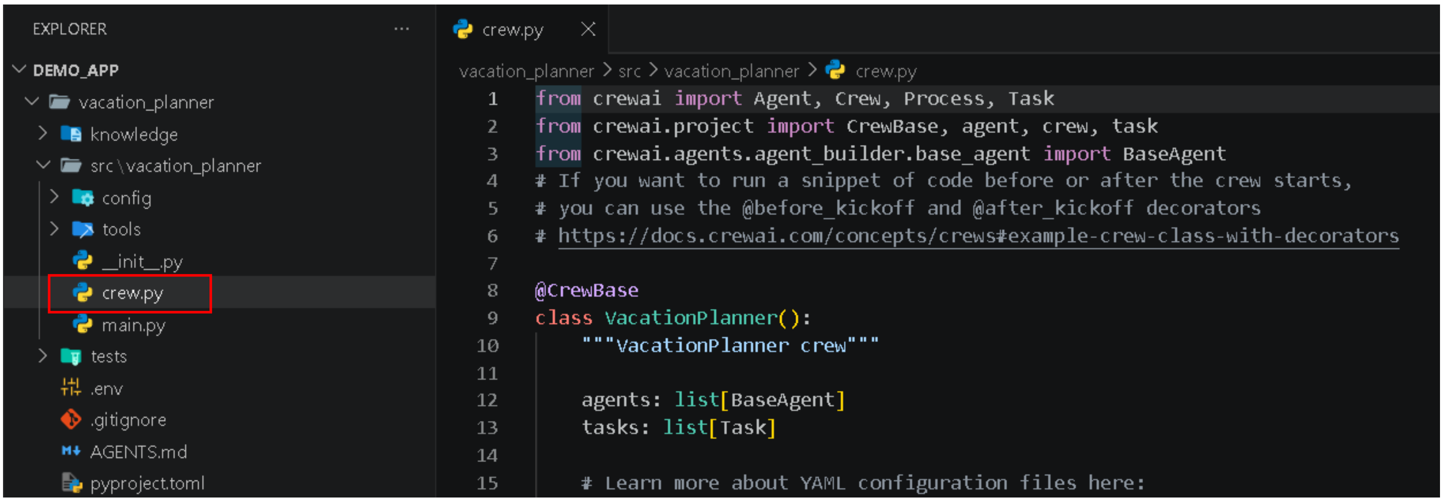
- Now Enter Details and click Create account.
- Now simply go to your Mail and complete verification.
- And then try to login.



- Now your account is created and login successfully to your account.
- Now In Navigation bar click API keys.
- And then simply copy it.
- Now go to your VS Code.



- Now open .env file inside in vacation_planner.
- And add SERPER_API_KEY = paste your API that you copied from serper website.



- Now click src.
- In src click vacation_planner.
- In vacation _planner open crew.py

crew.py 4 × agents.yaml

vacation_planner > src > vacation_planner > crew.py > VacationPlanner > vacation_researcher

```python
from crewai import Agent, Crew, Process, Task
from crewai.project import CrewBase, agent, crew, task
from crewai.agents.agent_builder.base_agent import BaseAgent
from crewai_tools import SerperDevTool
# If you want to run a snippet of code before or after the crew starts,
# you can use the @before_kickoff and @after_kickoff decorators
# https://docs.crewai.com/concepts/crews#example-crew-class-with-decorators

@CrewBase
class VacationPlanner():
    """VacationPlanner crew"""
```

- Now import SerperDevTool.
- And scroll down.

```python
21      # https://docs.crewai.com/concepts/agents#agent-tools
22      @agent
23      def vacation_researcher(self) -> Agent:
24          return Agent(
25              config=self.agents_config['vacation_researcher'], # type: ignore[index]
26              verbose=True,
27              tools=[SerperDevTool()],
28          )
29
```

- Now in line 22 first agent decorator(@agent).
- Copy first agent name from agents.yaml file.
- In agents.yaml first agent name that will search on internet is vacation_researcher
- copy this name from agents.yaml file and paste in line 23 and line no 24.
- Then add tools = [serperDevTool()].
- Now one thing we need to do is that define from where they will take API key.
- And we know that we define our API key in .env file.
- Basically we need to import one more thing and write one line code.
- So scroll up.

crew.py 4 × .env agents.yaml

vacation_planner > src > vacation_planner > crew.py > ...

```python
from crewai import Agent, Crew, Process, Task
from crewai.project import CrewBase, agent, crew, task
from crewai.agents.agent_builder.base_agent import BaseAgent
from crewai_tools import SerperDevTool
import os

#initialize SerperDevTool
serper_dev_tool = SerperDevTool(api_key=os.environ.get("SERPER_API_KEY"))

```

- Write these two code so they can access API key.

```python
    # https://docs.crewai.com/concepts/agents#agent-tools
    @agent
    def vacation_researcher(self) -> Agent:
        return Agent(
            config=self.agents_config['vacation_researcher'], # type: ignore[index]
            verbose=True,
            tools=[serper_dev_tool],
        )
```

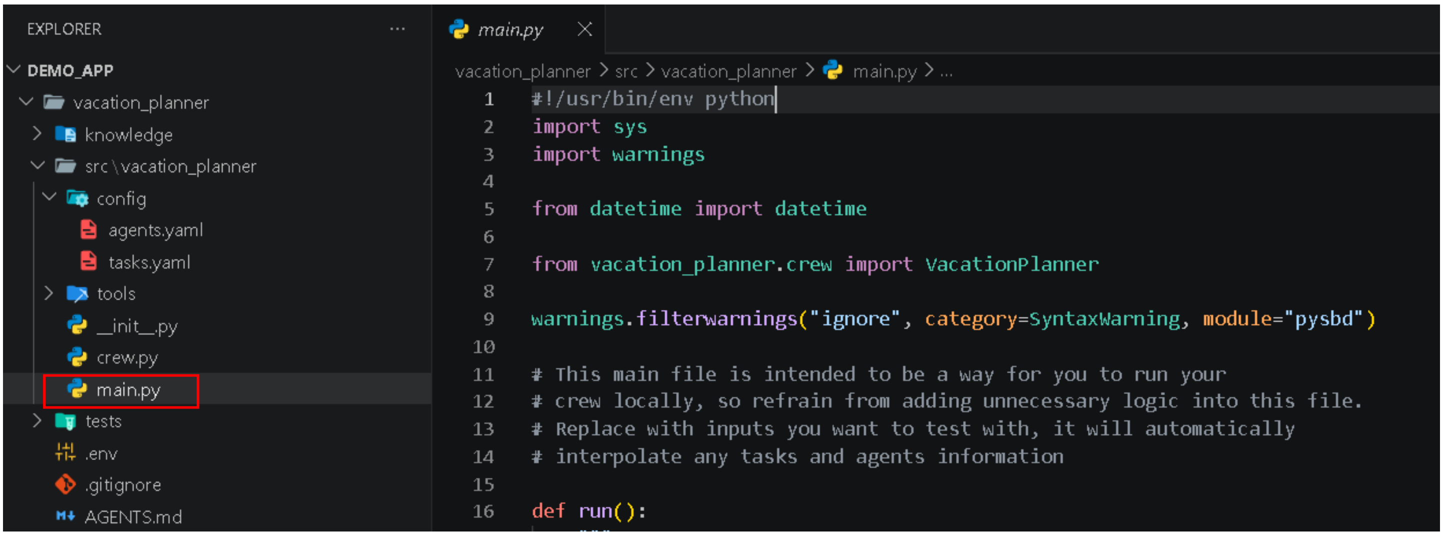
- Now go to first agent decorator (@agent).
- And in tools = [give that name you initialize in SerperDevToll].

```python
36      @agent
37      def itinerary_planner(self) -> Agent:
38          return Agent(
39              config=self.agents_config['itinerary_planner'], # type: ignore[index]
40              verbose=True
41          )
42
```

- Now scroll down and in line 36 you see 2nd agent decorator(@agent).
- And change its name with 2nd agent name that will present in agents.yaml file.
- Now in our case 2nd agent name is itinerary_planner
- Now replace in line 37 and line 39
- Now scroll down.

```python
    @task
    def research_task(self) -> Task:
        return Task(
            config=self.tasks_config['research_task'], # type: ignore[index]
        )

    @task
    def reporting_task(self) -> Task:
        return Task(
            config=self.tasks_config['reporting_task'], # type: ignore[index]
            output_file='report.md'
        )
```

- Now check tasks name from tasks.yaml file
- In our case we created two task in tasks.yaml file.
- So make sure in code task name is same as tasks.yaml file.
- Now all work donw in this file save this file,

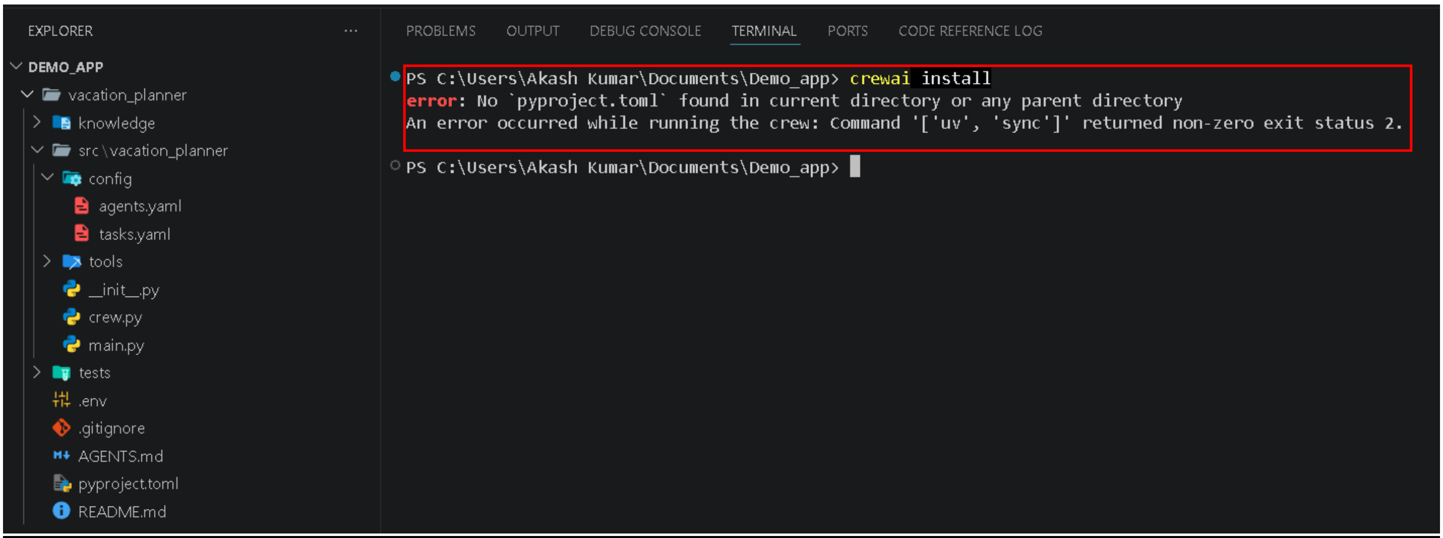


- Now in same src, vacation_planner file.
- Open main.py file.

- Now scroll down and in run function.
- Line no 21 give input topic.
- In our case our user want to know about London.
- So we give London.
- And save this file.

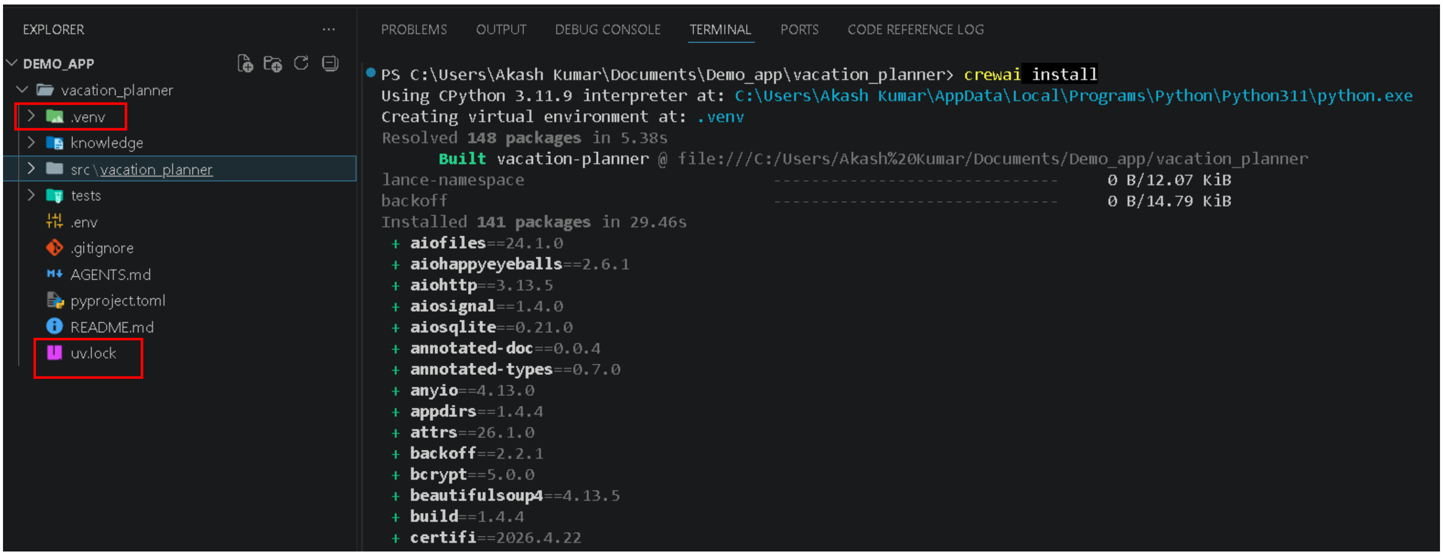
<u>Now run the Project: -</u>

- Open Terminal in VS Code.

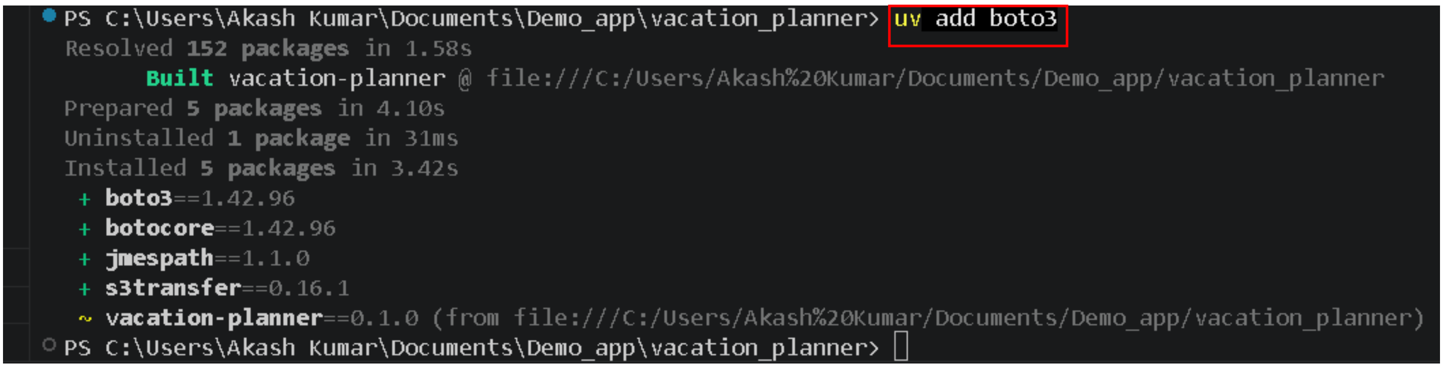


- Now you see it's give error.
- Because we are not in correct folder we are in Demo_app.
- But we need to go inside Demo_app and then inside vacation_planner that present in Demo_app.

- So simply we write cd vacation_planner
- And then lock the dependencies by this code crewai install.



- Now dependencies is locked and some new file also added.
- You see in left navigation bar.
- Now we need to install boto3 because its required for Amazon Bedrock.



- Our boto3 is installed.
- Now run the app.

```
PS C:\Users\Akash Kumar\Documents\Demo_app\vacation_planner> crewai run
Running the Crew
──────────────────────── 🚀 Crew Execution Started ────────────────────────

  Crew Execution Started
  Name: VacationPlanner
  ID: df40e067-8bf3-4290-b830-0789c61a8060

─────────────────────────────── 📋 Task Started ───────────────────────────────

  Task Started
  Name: research_task
  ID: 90b177cb-77af-41fa-9d5f-404fedb987a3

─────────────────────────────── 🤖 Agent Started ───────────────────────────────

  Agent: Senior Vacation Researcher

  Task: Conduct thorough research about the travel destination London. Find interesting and relevant information to
  enrich the travel experience.

──────────────────────── 🔧 Tool Execution Started (#1) ────────────────────────

  Tool: search_the_internet_with_serper
  Args: {'search_query': 'interesting and exciting facts about London'}
```

- You see first vacation planner Execution is started that search on Internet and give result using tool serper.

```
──────────────────────────── ✅ Agent Final Answer ────────────────────────────

  Agent: Senior Vacation Researcher

  Final Answer:
  - London Underground, opened in 1863, is the world's oldest subway system.
  - Three of the top ten museums and galleries in the world are in London, with a total of 857 art galleries.
  - London has four UNESCO World Heritage sites: Tower of London, Westminster, Greenwich, and Maritime Greenwich.
  - The City of London is the smallest city in the UK.
  - London has 20 secret underground rivers.
  - The Knowledge of London test is one of the hardest taxi knowledge tests in the world.
  - London is globally renowned for its rich history, vibrant culture, and landmarks like Tower Bridge, the Tower of
  London, Buckingham Palace, and the London Eye.
  - London is full of rolling green parks, cherry red buses, royal palaces, and is a multicultural city with diverse
  attractions.
  - London has ancient Roman roots and is home to the capital's oldest hat shop.
  - London's smallest public statue is of a mouse.
  - London is the world's largest financial center.
  - There are 13 ghosts haunting one of London's spookiest locations.
  - It is impossible for most people to lick their own elbow.
  - A crocodile cannot stick its tongue out.
  - A shrimp's heart is in its head.

───────────────────────────── 📋 Task Completion ─────────────────────────────

  Task Completed
  Name: research_task
  Agent: Senior Vacation Researcher
```

- Now you see 1st agent research on internet and gives result.
- 1st agent task is completed.

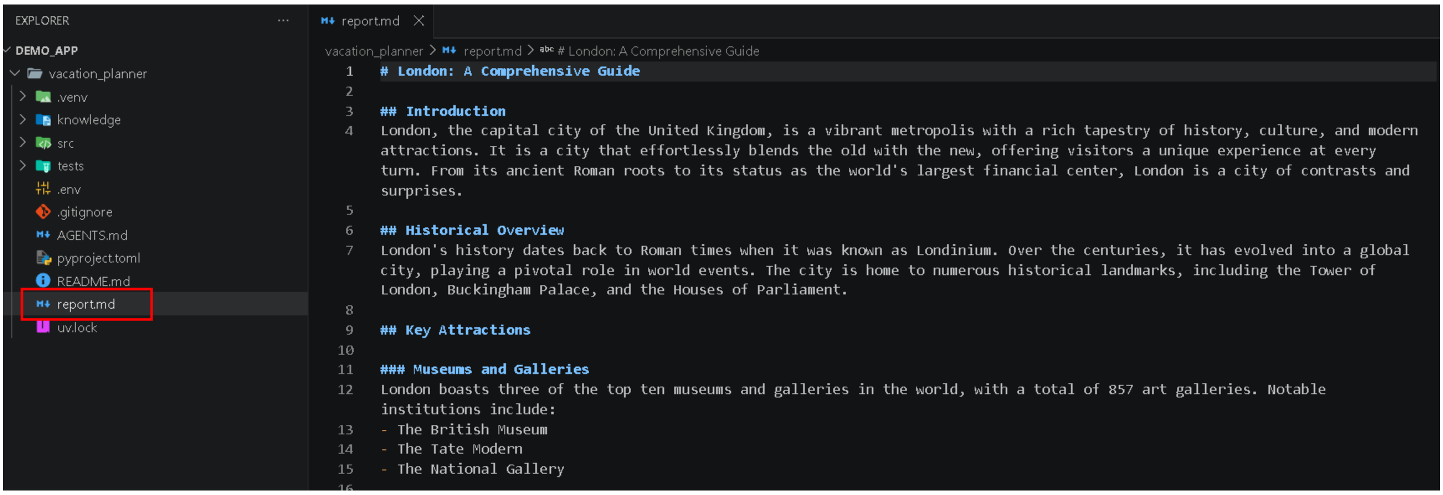
```
📋 Task Started

Task Started
Name: reporting_task
ID: 020d9c25-7e4a-43ea-b4e7-c0e04ed9416b
```

```
🤖 Agent Started

Agent: Itinerary Planner

Task: Review the research findings and create a detailed report about London.
```

- Now second agent and second task is executing.
- Basically it will review first agent result and give detailed report.

```
✅ Agent Final Answer

Agent: Itinerary Planner

Final Answer:
# London: A Comprehensive Guide

## Introduction
London, the capital city of the United Kingdom, is a vibrant metropolis with a rich tapestry of history, culture,
and modern attractions. It is a city that effortlessly blends the old with the new, offering visitors a unique
experience at every turn. From its ancient Roman roots to its status as the world's largest financial center, London
is a city of contrasts and surprises.

## Historical Overview
London's history dates back to Roman times when it was known as Londinium. Over the centuries, it has evolved into a
global city, playing a pivotal role in world events. The city is home to numerous historical landmarks, including
the Tower of London, Buckingham Palace, and the Houses of Parliament.

## Key Attractions

### Museums and Galleries
London boasts three of the top ten museums and galleries in the world, with a total of 857 art galleries. Notable
institutions include:
- The British Museum
- The Tate Modern
- The National Gallery

### UNESCO World Heritage Sites
London is home to four UNESCO World Heritage sites:
1. **Tower of London** - A historic castle located on the north bank of the River Thames.
2. **Westminster** - Includes Westminster Abbey, the Houses of Parliament, and Saint Margaret's Church.
3. **Greenwich** - Known for the Royal Observatory.
4. **Maritime Greenwich** - Features the Queen's House, the Royal Observatory, and the National Maritime Museum.

### Iconic Landmarks
- **Tower Bridge** - A combined bascule and suspension bridge built between 1886 and 1894.
- **Buckingham Palace** - The London residence and administrative headquarters of the monarch of the United Kingdom.
- **London Eye** - A giant Ferris wheel situated on the South Bank of the River Thames, offering panoramic views of
the city.
```

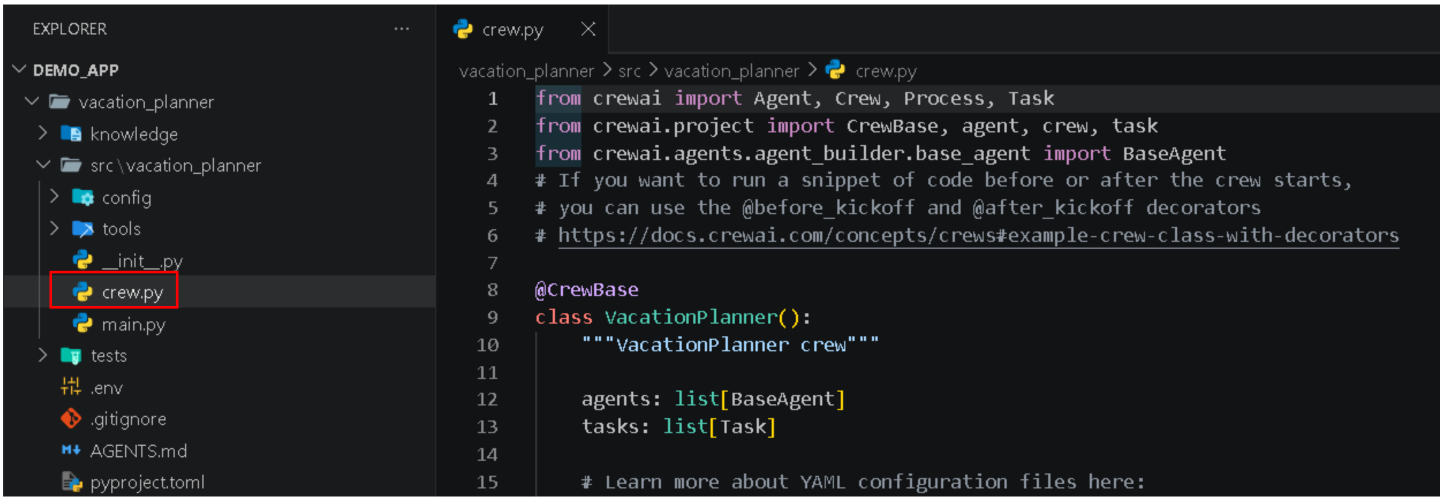
- And now this is the Second agent result.
- And then you scroll down you will see crew completion with all result.
- Now our both agent is working perfectly.



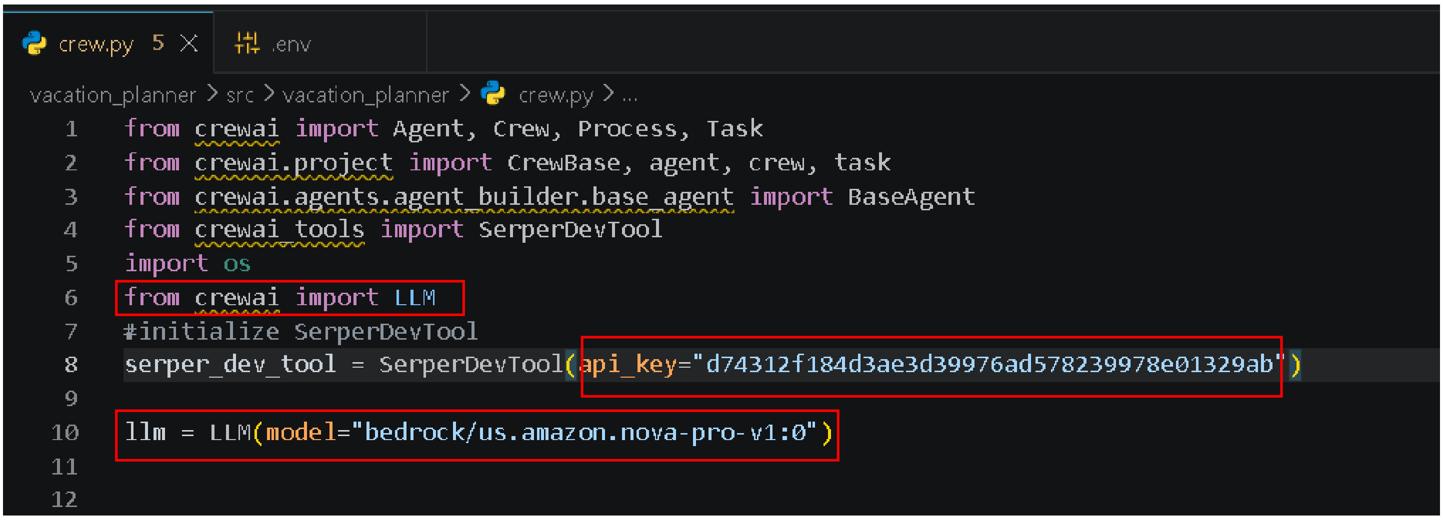
- Now in our file one report.md is created.
- Go and open it .
- Then you see there all details about that place you want to plan vacation are saved.

<u>Frontend: -</u>

- Now for Frontend we need to add one more thing in our code.
- Let's See.



- Open crew.py inside src vacation_planner folder.

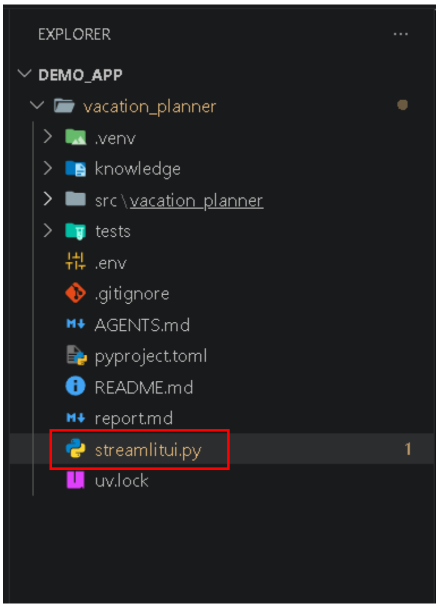


- Import LLM.
- In server_dev_tool that you write previously.
- Now replace api_key with default api key value that you paste in .env file.

- Now create llm = LLM(and copy model from .env file)

```python
    @agent
    def vacation_researcher(self) -> Agent:
        return Agent(
            config=self.agents_config['vacation_researcher'], # type: ignore[index]
            verbose=True,
            tools=[serper_dev_tool],
            llm=llm
        )

    @agent
    def itinerary_planner(self) -> Agent:
        return Agent(
            config=self.agents_config['itinerary_planner'], # type: ignore[index]
            verbose=True,
            llm=llm
        )
```

- Now in both agent decorators.
- add llm = llm.




```python
import streamlit as st
import os
from datetime import datetime
from src.vacation_planner.crew import VacationPlanner # Replace path 1

st.set_page_config(page_title="Personal AI Vacation Planner ", page_icon="✈️")

# Enhanced CSS styling
st.markdown("""
<style>
.main-header {
    background: linear-gradient(90deg, #FF6B6B, #4ECDC4, #45B7D1, #96CEB4);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
    font-size: 3rem;
    font-weight: bold;
    text-align: center;
    margin-bottom: 0;
}
```

- Now add this streamlitui.py in this folder (provided in resource).
- Now we need to install streamlit
- Open terminal and write code.

```
PS C:\Users\Akash Kumar\Documents\Demo_app> cd vacation_planner
PS C:\Users\Akash Kumar\Documents\Demo_app\vacation_planner> uv add streamlit
Resolved 168 packages in 4.69s
      Built vacation-planner @ file:///C:/Users/Akash%20Kumar/Documents/Demo_app/vacation_planner
Prepared 15 packages in 19.03s
Uninstalled 1 package in 1.41s
Installed 15 packages in 16.48s
 + altair==6.1.0
 + blinker==1.9.0
 + cachetools==7.0.6
 + gitdb==4.0.12
 + gitpython==3.1.47
 + narwhals==2.20.0
 + pandas==3.0.2
 + pydeck==0.9.2
 + smmap==5.0.3
 + streamlit==1.56.0
 + toml==0.10.2
 + tornado==6.5.5
 + tzdata==2026.2
 ~ vacation-planner==0.1.0 (from file:///C:/Users/Akash%20Kumar/Documents/Demo_app/vacation_planner)
 + watchdog==6.0.0
PS C:\Users\Akash Kumar\Documents\Demo_app\vacation_planner>
```

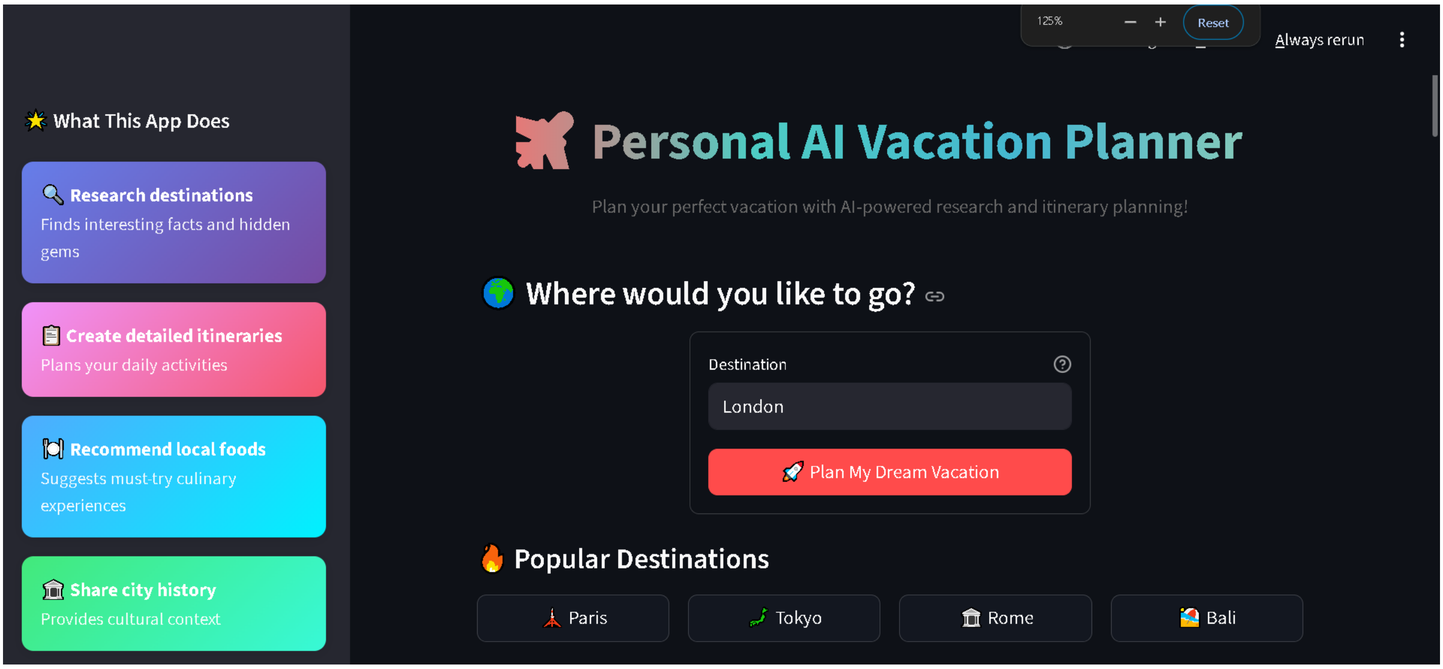
- Streamlit is installed
- Now simply run your streamlit.py file

```
PS C:\Users\Akash Kumar\Documents\Demo_app\vacation_planner> .\.venv\Scripts\activate
(vacation-planner) PS C:\Users\Akash Kumar\Documents\Demo_app\vacation_planner> pip install crewai crewai-tools streamlit
Collecting crewai
  Using cached crewai-1.14.3-py3-none-any.whl.metadata (36 kB)
Collecting crewai-tools
  Using cached crewai_tools-1.14.3-py3-none-any.whl.metadata (11 kB)
Collecting streamlit
  Using cached streamlit-1.56.0-py3-none-any.whl.metadata (9.8 kB)
Collecting aiofiles~=24.1.0 (from crewai)
  Using cached aiofiles-24.1.0-py3-none-any.whl.metadata (10 kB)
Collecting aiosqlite~=0.21.0 (from crewai)
```

- nNow activate virtual environment.
- Then install crewai tools and streamlit.

```
(vacation-planner) PS C:\Users\Akash Kumar\Documents\Demo_app\vacation_planner> python -m streamlit run streamlitui.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.31.238:8501
```

- Now run this by python –m streamlit run filename.
- Now open local URL in Browser.



- Now see its working perfectly.

- And gives results.